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CDC: a Python package for Color Distance Calculations

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Software

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Summary

The Color Distance Calculator (CDC) is an open-source python package designed to calculate a color distance image, a gray scale image with color distances from all pixels in the input image to a reference color. It is specifically tailored for handling large orthomosaics and multispectral data. By providing CDC with reference pixels, it calculates the distance using either the Mahalanobis distance (Wikipedia contributors, 2025) or a Gaussian Mixture Model for all pixels in the orthomosaic.

Figure 1 shows a small section of a pumpkin field and the calculated color distance image. CDC's main functions are exposed through a command-line interface, where providing an orthomosaic, a reference image, and a mask will output a new orthomosaic with the color distances. The Python package also allows for using CDC as a library for more complex tasks.

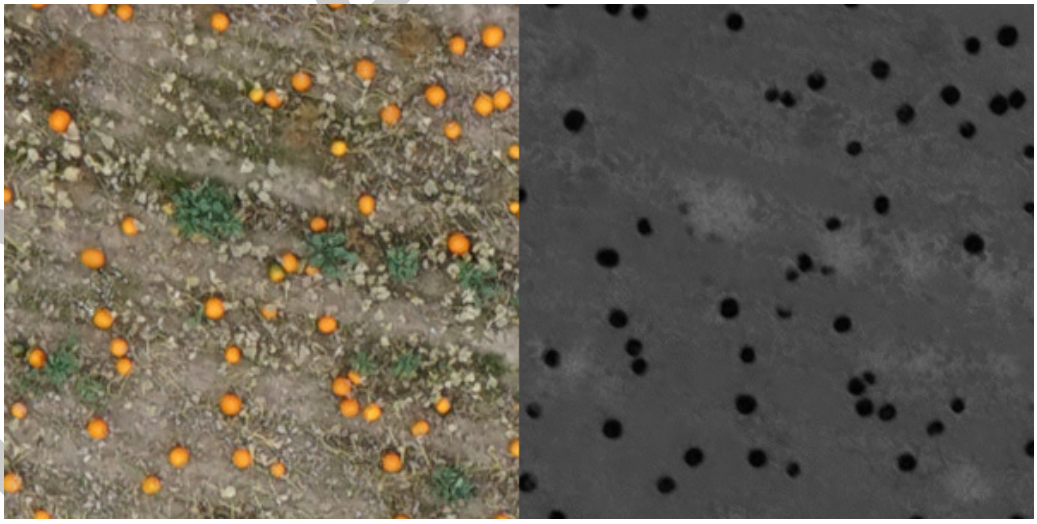

Figure 1: Small section of pumpkins field (left) color distance image of pumpkins field (right)

Statement of need

A common task in Precision Agriculture is to segment an orthomosaic into different regions based on the information in the orthomosaic. The regions can represent areas with healthy vegetation or areas with unwanted vegetation.

The classic approach is to use the excess green (ExG) (Woebbecke et al., 1995) color index to assess whether the current pixel is green enough to be considered healthy vegetation. Such

an approach based on a hardcoded, rule (ExG and a threshold) is only suitable for a limited number of cases.

Given enough training data, it is possible to train convolutional neural networks (CNN's) for segmenting arbitrary objects in images (Ronneberger et al., 2015). However, obtaining enough annotated training data can be difficult.

A more flexible approach is to use a small set of pixels to determine a reference color and then calculate the distance for all pixels in the input image to that reference color. We have successfully used this approach in several cases, including the following:

- detect healthy crop plants in a grass seed field
- locating thistles in a grass seed field
- counting pumpkins in a pumpkin field (Midtiby & Pastucha, 2022)

We propose CDC for segmenting large multispectral orthomosaics by calculating the color distance to a set of reference pixels. The Output of CDC is a grayscale orthomosaic which can easily be threshold to achieve a black and white segmentation.

CDC is developed with Agriculture uses in mind, but can easily be applied to other domains as is or by utilizing the library for custom needs.

Background and methods

In Precision Agriculture a common application is to assess the vegetation health by using Remote Sensing techniques and image analytics. The most applied Remote Sensing techniques is arial monitoring, where images from satellites, manned aircraft, and Unmanned Aerial Vehicles (UAVs) are captured (Matese et al., 2015). The use of UAVs, also known as drones, has seen a large increase in recent years as they provide high-quality images at a more affordable cost than satellites and manned aircraft (Radoglou-Grammatikis et al., 2020).

UAVs can carry various kinds of cameras, such as multispectral and hyperspectral, along with normal RGB cameras, thereby acquiring aerial images that can be used to extract vegetation indices. Vegetation indices such as the Normalized Difference Vegetation Index (NDVI) can be interpreted by farmers to monitor the crops variability and stress(Xue & Su, 2017). Individual images from the UAV normally only cover a small part of the field, but to get a overview of the whole field, the images are stitched together in software like OpenDroneMap, Metashape, or Pix4D together with Geographical Information Systems (GIS) information, creating a large georeferenced orthomosaic.

In CDC the Mahalanobis distance in the RGB color space is used as the default color distance. The Mahalanobis distance is defined as

$$\sqrt{(\vec{x}-\vec{\mu})^T \cdot S^{-1} \cdot (\vec{x}-\vec{\mu})}$$

where $\vec{x}$ is the color value, $\vec{\mu}$ the mean color value and S the covariance matrix. The parameters $\vec{\mu}$ and S are determined from a set of training pixels. If the color value distribution have multiple peaks, it can be beneficial to use the Gaussian Mixture Model from the Scikit Learn (Pedregosa et al., 2011) python package.

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